IV Bolus Practice 1

Module 2 - IV Bolus Administration · Daily practice sheet · No date

Name _____ Date _____

a The data presented in the table below are plasma concentrations of a new drug at different time points after an intravenous bolus injection of 50 mg to a 70-kg patient.

Time (hr)	Plasma concentration (mg/L)
0.5	2.52
1.0	1.59
1.5	1.00
2.0	0.64
2.5	0.40
3.0	0.25

What is the elimination rate constant?

b A drug given by intravenous bolus injection has an elimination rate constant of 0.91 hr^{-1}. What is its half-life of elimination?

c After an intravenous bolus injection of 50 mg, a drug with an elimination rate constant of 0.91 hr^{-1} has a plasma concentration of 0.40 mg/L at 2.5 hours. What is the initial plasma concentration?

d A drug given by intravenous bolus injection has an initial plasma concentration of 3.89 mg/L and an elimination rate constant of 0.91 hr^{-1}. What is the concentration of drug in the plasma 15 minutes after the dose was given?

e A 50 mg intravenous bolus dose produces an initial plasma concentration of 3.89 mg/L in a 70-kg patient. What is the apparent volume of distribution?

f A drug has an apparent volume of distribution of 12.85 L and an elimination rate constant of 0.91 hr^{-1} in a patient. What is the total body clearance of this drug in this patient?

g A 50 mg intravenous bolus dose of a drug with a volume of distribution of 12.85 L gives a plasma concentration of 0.25 mg/L at 3 hours. What is the amount of drug in the body 3 hours after the drug was administered?

h A drug given by intravenous bolus injection has an elimination half-life of 0.76 hours. How much time is required for 99.9% of the drug to be eliminated from the body?